

24 DSA PATTERNS

+ 3 LeetCode Questions Per Pattern

No 20 playlists. No 500 random questions.

Just the patterns that show up again and again at Swiggy, Razorpay, Flipkart, Zepto, PhonePe - plus 3 hand-picked LeetCode problems per pattern to drill it.

HOW TO USE THIS SHEET

1. Pick one pattern a day. Solve all 3 questions before moving on.
2. Don't jump patterns mid-week - depth beats breadth here.
3. Second pass: mix patterns randomly to simulate real interviews.

THE PATTERNS

01 Two Pointers

When: Sorted array / palindrome problems

- Two Sum II - Input Array Is Sorted
<https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/>
- Container With Most Water
<https://leetcode.com/problems/container-with-most-water/>
- 3Sum
<https://leetcode.com/problems/3sum/>

02 Sliding Window

When: Subarray/substring with a condition

- Longest Substring Without Repeating Characters
<https://leetcode.com/problems/longest-substring-without-repeating-characters/>
- Minimum Window Substring
<https://leetcode.com/problems/minimum-window-substring/>
- Longest Repeating Character Replacement
<https://leetcode.com/problems/longest-repeating-character-replacement/>

03 Fast & Slow Pointers

When: Cycle / midpoint detection

- Linked List Cycle
<https://leetcode.com/problems/linked-list-cycle/>
- Linked List Cycle II
<https://leetcode.com/problems/linked-list-cycle-ii/>
- Middle of the Linked List
<https://leetcode.com/problems/middle-of-the-linked-list/>

04 Merge Intervals

When: Overlapping ranges

- Merge Intervals
<https://leetcode.com/problems/merge-intervals/>
- Insert Interval
<https://leetcode.com/problems/insert-interval/>
- Non-overlapping Intervals
<https://leetcode.com/problems/non-overlapping-intervals/>

05 Binary Search

When: Sorted search space

- Binary Search
<https://leetcode.com/problems/binary-search/>
- Search in Rotated Sorted Array
<https://leetcode.com/problems/search-in-rotated-sorted-array/>
- Find Minimum in Rotated Sorted Array
<https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/>

THE PATTERNS

06 Binary Search on Answer

When: Minimize/maximize a feasible value

- Koko Eating Bananas
<https://leetcode.com/problems/koko-eating-bananas/>
- Capacity To Ship Packages Within D Days
<https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/>
- Split Array Largest Sum
<https://leetcode.com/problems/split-array-largest-sum/>

07 DFS (Tree/Graph)

When: Explore all paths/depth

- Number of Islands
<https://leetcode.com/problems/number-of-islands/>
- Path Sum
<https://leetcode.com/problems/path-sum/>
- Max Area of Island
<https://leetcode.com/problems/max-area-of-island/>

08 BFS (Tree/Graph)

When: Shortest path / level order

- Rotting Oranges
<https://leetcode.com/problems/rotting-oranges/>
- Word Ladder
<https://leetcode.com/problems/word-ladder/>
- Binary Tree Level Order Traversal
<https://leetcode.com/problems/binary-tree-level-order-traversal/>

09 Backtracking

When: All combinations/permutations

- Subsets
<https://leetcode.com/problems/subsets/>
- Permutations
<https://leetcode.com/problems/permutations/>
- N-Queens
<https://leetcode.com/problems/n-queens/>

10 DP - 1D

When: Overlapping subproblems, linear state

- Climbing Stairs
<https://leetcode.com/problems/climbing-stairs/>
- House Robber
<https://leetcode.com/problems/house-robber/>
- Longest Increasing Subsequence
<https://leetcode.com/problems/longest-increasing-subsequence/>

THE PATTERNS

11 DP - 2D / Grid

When: Grid paths, string matching

- Unique Paths
<https://leetcode.com/problems/unique-paths/>
- Minimum Path Sum
<https://leetcode.com/problems/minimum-path-sum/>
- Edit Distance
<https://leetcode.com/problems/edit-distance/>

12 DP - Knapsack

When: Choose/skip with a constraint

- Partition Equal Subset Sum
<https://leetcode.com/problems/partition-equal-subset-sum/>
- Coin Change
<https://leetcode.com/problems/coin-change/>
- Target Sum
<https://leetcode.com/problems/target-sum/>

13 Greedy

When: Locally optimal choice

- Jump Game
<https://leetcode.com/problems/jump-game/>
- Gas Station
<https://leetcode.com/problems/gas-station/>
- Candy
<https://leetcode.com/problems/candy/>

14 Topological Sort

When: Ordering with dependencies

- Course Schedule
<https://leetcode.com/problems/course-schedule/>
- Course Schedule II
<https://leetcode.com/problems/course-schedule-ii/>
- Find Eventual Safe States
<https://leetcode.com/problems/find-eventual-safe-states/>

15 Union Find

When: Connected components

- Number of Provinces
<https://leetcode.com/problems/number-of-provinces/>
- Redundant Connection
<https://leetcode.com/problems/redundant-connection/>
- Accounts Merge
<https://leetcode.com/problems/accounts-merge/>

THE PATTERNS

16 Heap / Priority Queue

When: Top-K / running min-max

- Kth Largest Element in an Array
<https://leetcode.com/problems/kth-largest-element-in-an-array/>
- Merge k Sorted Lists
<https://leetcode.com/problems/merge-k-sorted-lists/>
- Top K Frequent Elements
<https://leetcode.com/problems/top-k-frequent-elements/>

17 Trie

When: Prefix-based lookup

- Implement Trie (Prefix Tree)
<https://leetcode.com/problems/implement-trie-prefix-tree/>
- Word Search II
<https://leetcode.com/problems/word-search-ii/>
- Design Add and Search Words Data Structure
<https://leetcode.com/problems/design-add-and-search-words-data-structure/>

18 Monotonic Stack

When: Next greater/smaller element

- Daily Temperatures
<https://leetcode.com/problems/daily-temperatures/>
- Largest Rectangle in Histogram
<https://leetcode.com/problems/largest-rectangle-in-histogram/>
- Next Greater Element I
<https://leetcode.com/problems/next-greater-element-i/>

19 Monotonic Deque

When: Sliding window max/min

- Sliding Window Maximum
<https://leetcode.com/problems/sliding-window-maximum/>
- Shortest Subarray with Sum at Least K
<https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/>
- Constrained Subsequence Sum
<https://leetcode.com/problems/constrained-subsequence-sum/>

20 Prefix Sum

When: Range sum queries

- Subarray Sum Equals K
<https://leetcode.com/problems/subarray-sum-equals-k/>
- Range Sum Query - Immutable
<https://leetcode.com/problems/range-sum-query-immutable/>
- Product of Array Except Self
<https://leetcode.com/problems/product-of-array-except-self/>

THE PATTERNS

21 Bit Manipulation

When: XOR/bitmask tricks

- Single Number
<https://leetcode.com/problems/single-number/>
- Number of 1 Bits
<https://leetcode.com/problems/number-of-1-bits/>
- Counting Bits
<https://leetcode.com/problems/counting-bits/>

22 Linked List In-Place Ops

When: Reverse/reorder without extra space

- Reverse Linked List
<https://leetcode.com/problems/reverse-linked-list/>
- Reverse Linked List II
<https://leetcode.com/problems/reverse-linked-list-ii/>
- Reorder List
<https://leetcode.com/problems/reorder-list/>

23 Matrix Traversal

When: Spiral, rotate, in-place grid ops

- Rotate Image
<https://leetcode.com/problems/rotate-image/>
- Spiral Matrix
<https://leetcode.com/problems/spiral-matrix/>
- Set Matrix Zeroes
<https://leetcode.com/problems/set-matrix-zeroes/>

24 Recursion / Divide & Conquer

When: Break into subproblems

- Sort an Array (Merge Sort)
<https://leetcode.com/problems/sort-an-array/>
- Pow(x, n)
<https://leetcode.com/problems/powx-n/>
- Maximum Subarray
<https://leetcode.com/problems/maximum-subarray/>